

Data Science and Machine Learning

NLP III

Why are Language models so powerful?



Size Attention



mechanism



Layers



Etc.



Transfer Learning

Transfer Learning

What: *Store 'language knowledge' and 'task knowledge' in the parameters of a model.*

Why: *Learn new tasks faster & better.*

Classical Algorithms
(e.g. Regression, SVM)

No prior

‘language knowledge’

No knowledge of semantic similarities between words like "attack", "war" and ”tree".

No prior

‘task knowledge’

No knowledge of tasks like "Classify this text into ‘activist’ or ‘conservative’ rhetoric".

Word Embeddings
(e.g. Word2Vec)

Provide **'language knowledge'***:*

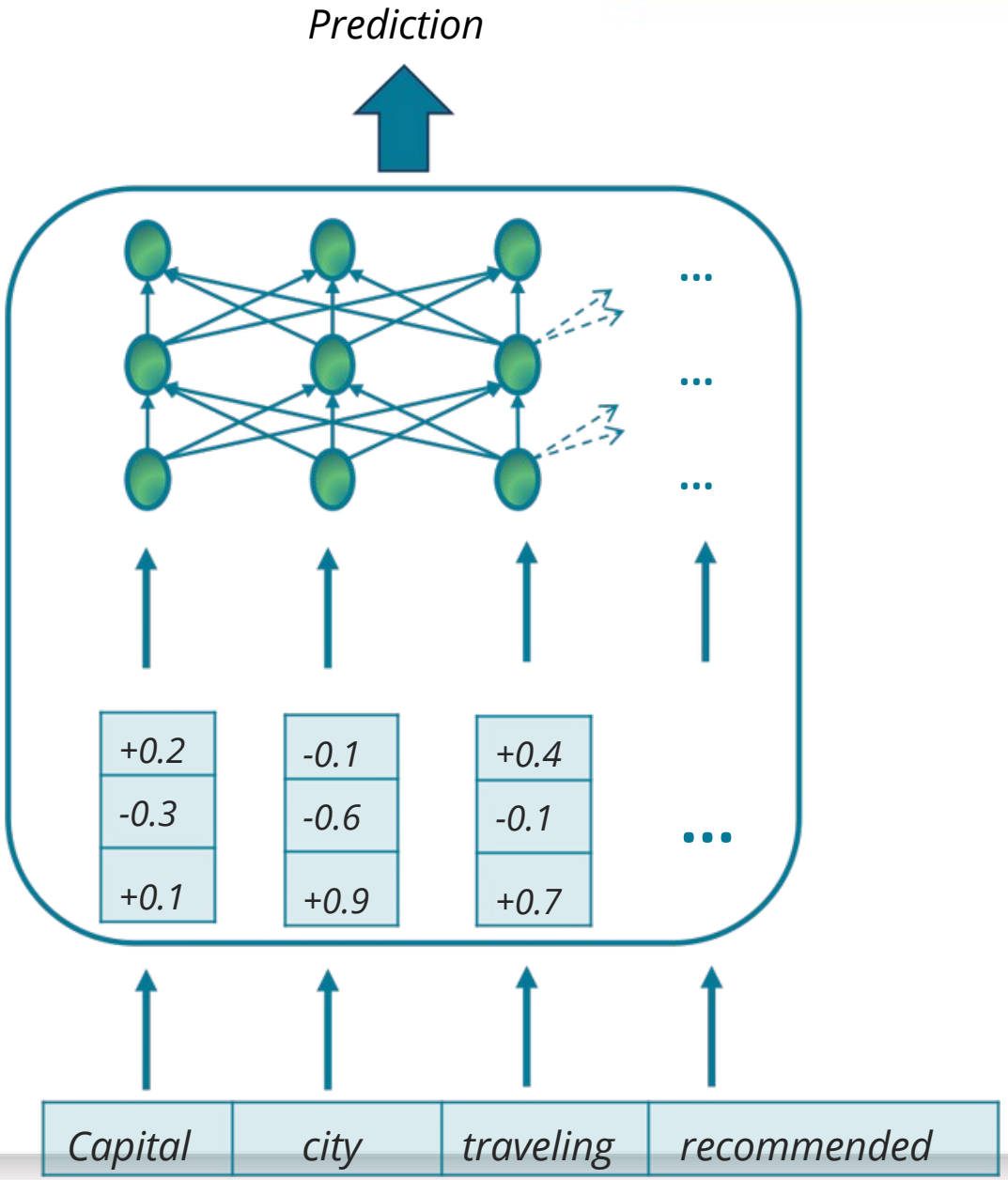
- *Represents "attack", "war" in similar static vectors*

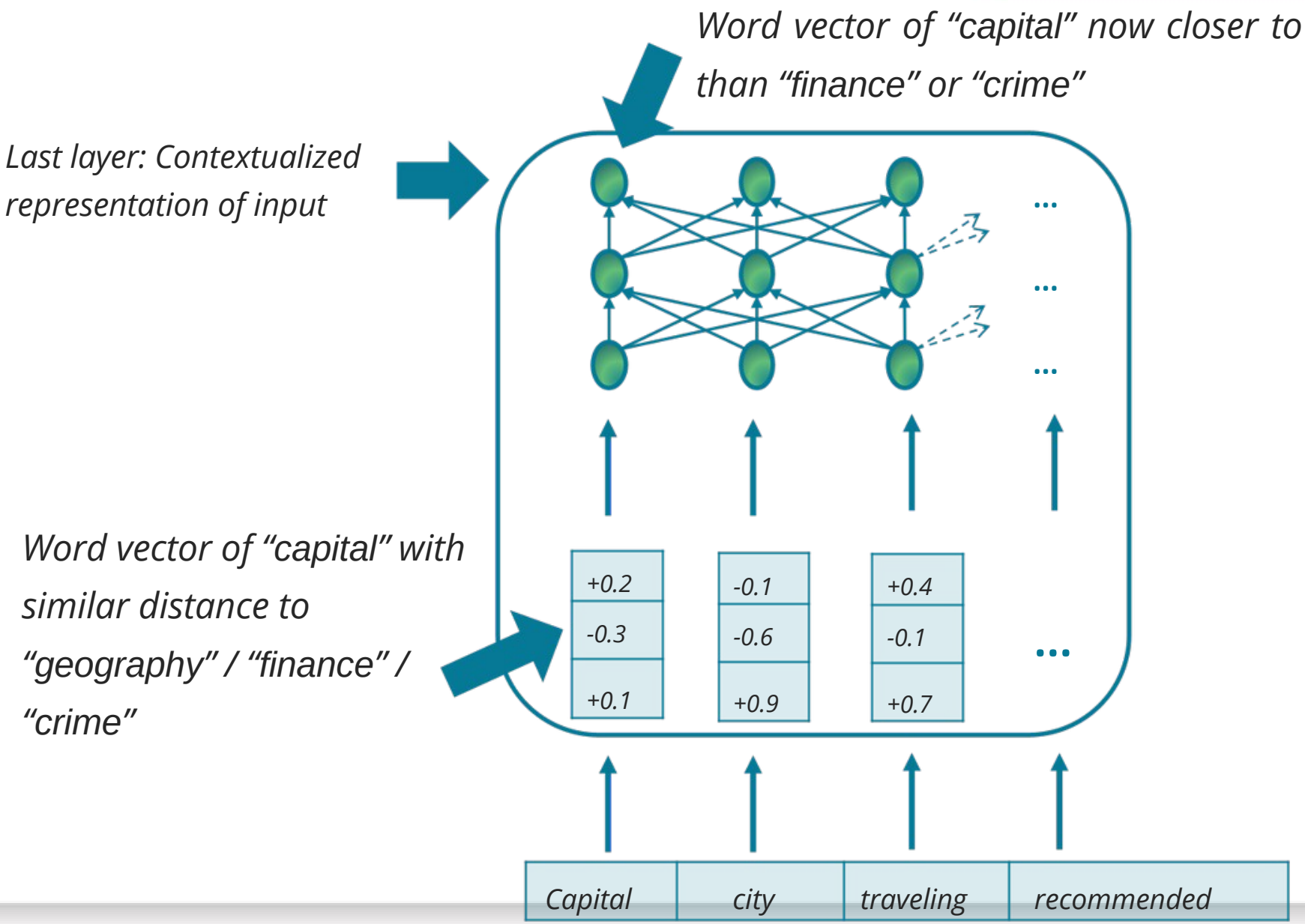
Attack	≈	*0.1*	*0.8*	*...*	*-0.5*	*0.1*
War	≈	*0.2*	*0.8*	*...*	*-0.4*	*0.3*
Tree	≈	*0.7*	*-0.4*	*...*	*0.1*	*-0.7*

Transformers
(e.g. BERT-base)

Prior **'language knowledge'***:*

- *Represents "attack", "war" in similar vectors*
- *Represents "capital" differently in context of "city" or "investment" or "crime"*


Prediction
+0.2
-0.3
+0.1
-0.1
-0.6
+0.9
+0.4
-0.1
+0.7
Capital
city
traveling
recommended


Word vector of “capital” now closer to “geography” than “finance” or “crime”
Last layer: Contextualized representation of input
Word vector of “capital” with similar distance to “geography” / “finance” / “crime”
+0.2
-0.3
+0.1
-0.1
-0.6
+0.9
+0.4
-0.1
+0.7
Capital
city
traveling
recommended

Transformers

(e.g. BERT-base)

Prior **'language knowledge***':*

- *Represents "attack", "war" in similar vectors*
- *Represents "capital" differently in context of "city" or*

  “investment” or “crime”

Learned through **simple, self-supervised task***:*

- *Masked Language Modelling*

How BERT acquires language knowledge

ORIGINAL TEXT

"Capital punishment, also known as thedeath penaltyand formerly calledjudicial homicide, is ..."

MASKED TEXT

"Capital **[MASK]**, *also known as thedeath* **[MASK]***and formerly called***[MASK]** *homicide, is ..."*

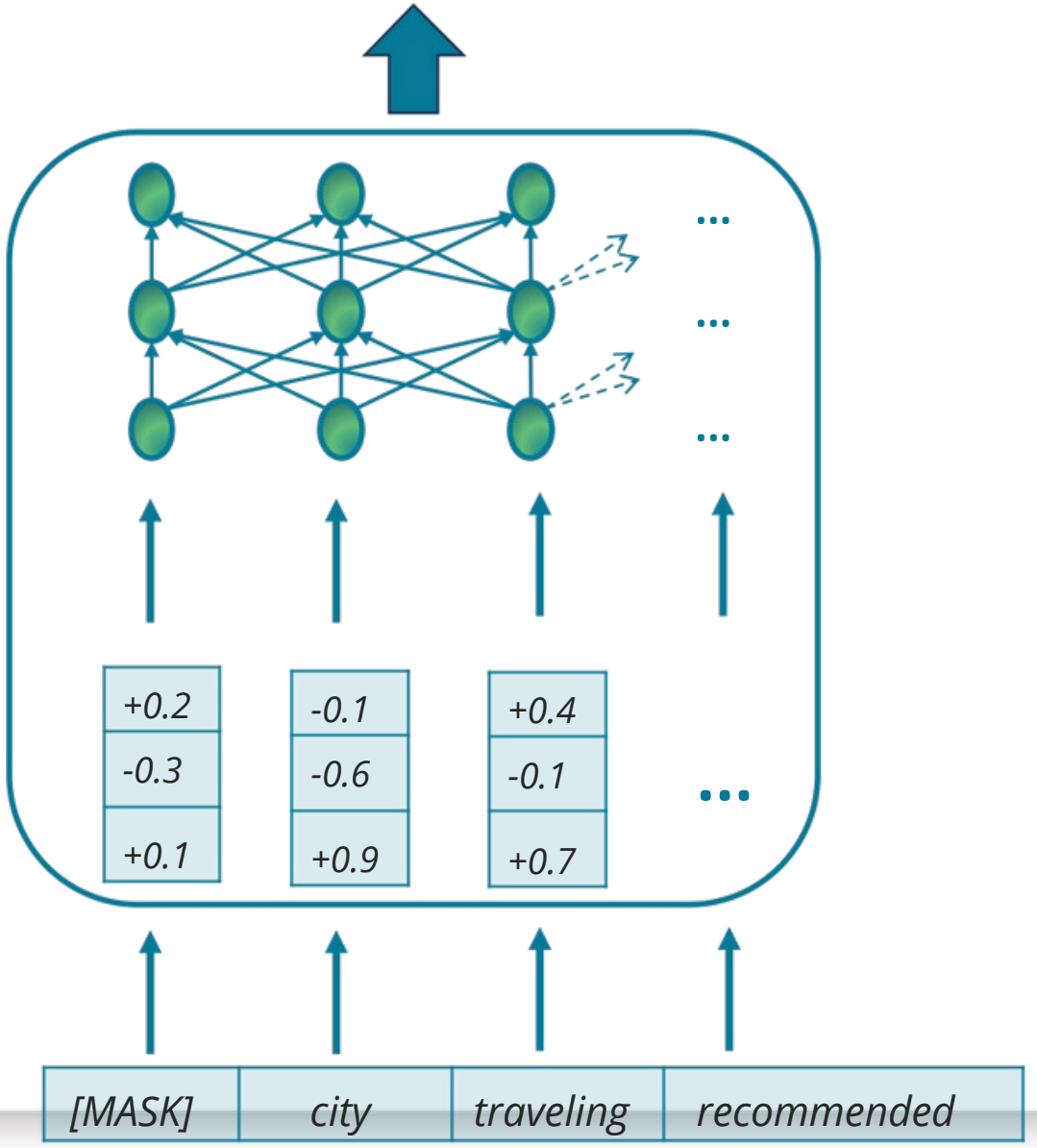
The algorithm learns to predict the correct word behind the [MASK] token.

This creates general **'language knowledge'**

BERT-base

After millions of training iterations:

Many parameters (vectors), which are very good at predicting hidden words.



Disadvantages of BERT-base

‘Task knowledge’ from MLM is not useful:

- *BERT-base only knows how to predict hidden words (MLM task)*
- *We are actually interested in other tasks like classification, summarization etc.*
- *BERT-base needs to learn new, useful tasks from scratch*

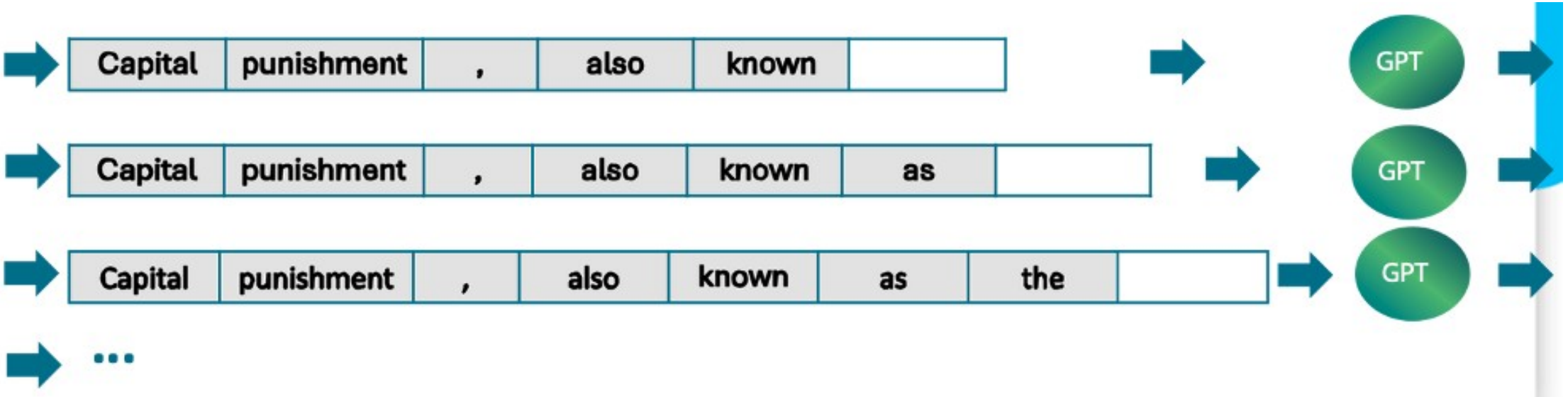
A more universal task

Next-token-prediction

- *Main pre-training task of GPT models*
- *Is a self-supervised task (no manual annotations)*

Next-token-prediction task

Original text: *“Capital punishment, also known as thedeath penaltyand formerly calledjudicial homicide, is ...”*



Universal task

Sentiment classification

[long text ...]	*Is*	*this*	*text*	*positive*	*or*
negative	*?*				



[long text ...]	*Is*	*this*	*text*	*positive*	*or*
negative	*?*	**positive**			

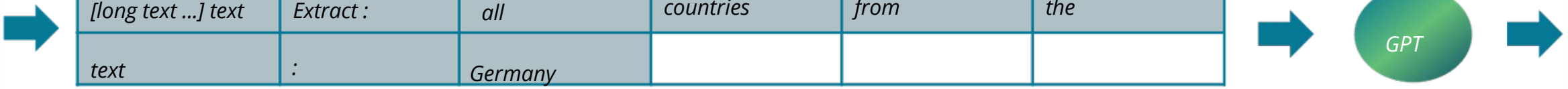
Universal task

Information extraction

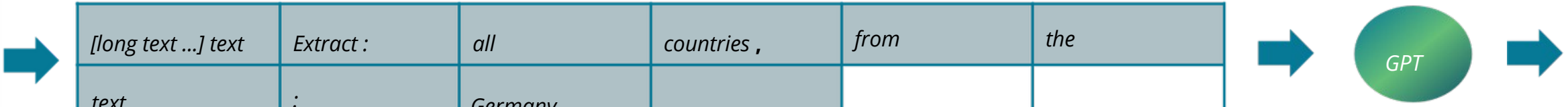
[long text …] text	Extract :	all	countries	from	the
text	:				



[long text …] text	Extract :	all	countries	from	the
text	:	Germany			



[long text …] text	Extract :	all	countries ,	from	the
text	:	Germany	,		



Universal task

Summarisation

[long text …] text	*A*	*summary*	*of*	*the*	*preceding*	
text	*:*					GPT

[long text …] text	*A*	*summary* **The**	*of*	*the*	*preceding*	
	:					GPT

[long text …] text	*A*	*summary* **The**	*of*	*the*	*preceding*	
	:		**main**			GPT



Reflect and Q&A

- *Q1: What is the difference between word embeddings and BERT?*
- *Q2: What are universal tasks and why are they useful?*
- *Q3: In your own words, try to define transfer learning.*

Write your responses on a piece of paper / notebook. Ask any questions about the slides in the chat.